# SUSHIN BANDHA

*Artificial Intelligence | Product Management | Project Management*

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## PROFESSIONAL SUMMARY

Purdue University AI student (B.S., Expected May 2027) with hands-on experience in product management, identity & access management (IAM), machine learning, and full-stack AI development. Licensed EMT (NREMT, New Jersey, and Indiana) actively practicing in inter-facility patient transport with 56+ hours of additional clinical field experience across hospital ER and pre-hospital EMS environments. Delivered a 99%-defect-free IAM platform launch at PSEG, led a 25-member cross-functional team, and built an MLOps pipeline reducing model update time by 97%. Google AI Professional Certificate holder, currently leading and contributing to independent and coursework technical projects spanning aviation systems, HCI research, and emergency-response platforms. Seeking IT, AI, or related roles where I can apply technical depth and a product-ownership mindset to build secure, scalable, intelligent systems.

## TECHNICAL SKILLS

**Languages & Frameworks:** Python, Java, C/C++, SQL, JavaScript, TypeScript, R, HTML, Bash/Linux, Agile/Scrum

**AI / ML:** PyTorch, TensorFlow, Scikit-learn, NLP, Deep Learning, Generative AI, RAG, LLM Fine-tuning, MLOps, Time-Series Forecasting

**Cloud & DevOps:** AWS, Vercel, CI/CD (GitHub Actions), Docker (familiar), Linux Systems Administration, Serverless Architecture

**Security & IAM:** Identity & Access Management (IAM), UAT, SIT, Access Control, Security Operations, Risk Assessment, Compliance Documentation

**Tools & Platforms:** PostgreSQL, Next.js, Salesforce, Excel, PowerPoint, Git, Vercel, Resend, Google API, Power Platform, Microsoft Azure, SAP, JIRA (familiar), Prompt Engineering, Ollama, Figma

**Medical & EMS:** BLS, NREMT Protocols, Patient Assessment, Vitals Monitoring, Patient Care Reporting (PCR), Inter-Facility Transport, Pre-Hospital Emergency Response

**Soft Skills:** Technical Leadership, Stakeholder Management, Cross-functional Collaboration, AI Ethics, Analytical Problem-Solving, Strategic Communication

## CERTIFICATIONS & LICENSES

**Google AI Professional Certificate** | Google / Coursera (7 courses) — *Completed May 2026*

**National Registry Emergency Medical Technician (NREMT)** | EMS ID: 3668-4546-3129 — *Certified Apr 2026 — Active*

**New Jersey Emergency Medical Technician (EMT) License** — *Active*

**Indiana Emergency Medical Technician (EMT) License** — *Active*

**FERPA — Family Educational Rights and Privacy Act** | Purdue University — *Completed Aug 2025 — Active*

**HIPAA — Health Insurance Portability and Accountability Act** | Purdue University — *Completed Aug 2025 — Active*

**GLBA — Gramm-Leach-Bliley / Financial Services Modernization Act** | Purdue University — *Completed Aug 2025 — Active*

**Data Classification and Handling Educational Resources** | Purdue University — *Completed Aug 2025 — Active*

**Protecting Social Security Numbers at Purdue University** | Purdue University — *Completed Aug 2025 — Active*

**Golf Cart Safety** | Purdue University — *Completed Aug 2025 — Active*

## EDUCATION

**B.S. Artificial Intelligence — Expected May 2027 | Purdue University — West Lafayette, IN** — *Aug 2023 – May 2027*

**Minor:** Computer Networking and Information Technology (CNIT)

**GPA:** 3.01 / 4.00

**Relevant Coursework:** Data Mining & Machine Learning, Data Structures & Algorithms, Systems Programming, Computer Architecture, OOP (Java), Programming in C, Human-Computer Interaction, Database Systems

**Activities:** BASF x The Data Mine (Machine Learning Practitioner); Floor Senator — Purdue University Residences (Aug 2025–May 2026)

## PROFESSIONAL EXPERIENCE

### OnTime Transport — NJ

**Emergency Medical Technician (EMT) | NJ** — *May 2026 – Present*

- Provide BLS patient care during inter-facility and non-emergency patient transports — conducting patient assessments, monitoring vital signs, and delivering structured clinical handoffs to receiving facilities.
- Apply NREMT-certified protocols and standard of care across varied patient acuity levels, ensuring patient safety, dignity, and continuity of care throughout the transport workflow.
- Document patient encounters with accurate, HIPAA-compliant Patient Care Reports (PCRs), and complete pre-shift rig and equipment readiness checks to maintain operational compliance.
- Collaborate with paramedics, nurses, and dispatch teams to coordinate timely transports and respond effectively to dynamic patient needs in time-sensitive environments.

### Purdue University Police Department — West Lafayette, IN

**PSSP Assistant Team Lead — Window Program | West Lafayette, IN** *Sep 2025 – Present*

- Lead administrative operations and day-to-day workflow management for the Purdue University Police Department's Window Program, overseeing front-desk operations, public-facing service requests, and coordination between civilian staff and sworn officers.
- Own end-to-end shift coverage and scheduling for the patroller team, balancing personnel availability against fluctuating staffing demands to maintain 100% coverage of critical posts.
- Spearheaded a process-improvement project to overhaul appointment-based fingerprinting operations — designed a staffing model and scheduling protocol that guarantees a trained patroller is on site for every scheduled appointment, eliminating service gaps and reducing missed appointments.
- Train and onboard incoming patrollers on department SOPs, front-desk protocols, and customer service standards — accelerating ramp-up and driving consistent service quality across all shifts.
- Serve as the primary escalation point between patrollers and full-time officers, triaging incoming requests, routing complex issues to the appropriate department personnel, and maintaining clear lines of communication across the department.
- Apply structured risk assessment frameworks to identify and mitigate environmental safety hazards, maintaining campus security compliance across academic and administrative assets.

### Purdue Athletics — West Lafayette, IN

**Event Supervisor — Event Operations & Security | West Lafayette, IN** *Aug 2025 – Present*

- Supervised multi-location event operations for Big Ten athletic events drawing 60,000+ patrons, coordinating cross-site staff deployment and managing real-time communication across security, gate, and parking teams.
- Directed cross-functional staff teams spanning multiple entry points and facility zones, delegating responsibilities, resolving personnel conflicts, and ensuring consistent service delivery under high-pressure conditions.
- Led customer support operations on the ground — de-escalating patron disputes, resolving ticketing and access issues, and serving as the primary point of contact for attendee concerns during live events.
- Executed time-critical incident response decisions, coordinating with law enforcement and facility management to resolve safety incidents swiftly while maintaining zero disruption to event flow.
- Applied data-driven resource allocation strategies based on lot capacity and permit-tier analytics, maximizing space utilization and reducing entry/exit congestion by 60% at peak-attendance events.

### PSEG — Newark, NJ

**Product Management Intern — Identity & Access Management (IAM) | Newark, NJ** *Jun 2025 – Aug 2025*

- Served as product owner for an enterprise IAM platform, triaging and prioritizing a 50+ item backlog of security bug fixes and feature enhancements, improving system usability and access-control reliability.
- Achieved 99% defect-free launch by designing and executing 300+ UAT and SIT test cases, validating end-to-end user workflows and access permission logic across business-critical systems.
- Built a business case for a new IAM scalability solution — conducted market analysis and stakeholder user research using Excel and PowerPoint — securing executive approval and a documented growth roadmap.
- Collaborated with security engineers, compliance leads, and business stakeholders using Agile practices to deliver iterative product improvements on schedule.

**Product Intern — Automation & Internal Tools | Newark, NJ** *Jun 2024 – Aug 2024*

- Engineered and presented 5 Proof of Concept (PoC) automation tools to senior leadership, securing executive approval and budget allocation for full-scale product development.
- Led a cross-functional team of 25 interns to define requirements, architect, and launch a company-wide employee onboarding portal — directly accelerating new-hire time-to-productivity.
- Owned end-to-end development of a critical process automation tool using Python and Salesforce, transforming a multi-day manual workflow into a near-instant execution pipeline.

## CLINICAL EXPERIENCE

**IU Health — Arnett Hospital | Lafayette, IN** *Fall 2026*

***Hospital Clinical Rotation | ~28 hours | Emergency Department***

- Completed a 28-hour emergency department clinical rotation, observing and assisting with EMT functions within a hospital setting across live ER intake and patient care workflows.
- Gained hands-on exposure to hospital-based emergency protocols, learning how EMT scope of practice and standard of care (SOC) differs between pre-hospital ambulance environments and in-hospital ER operations.
- Supported clinical staff during patient assessment, triage support, and care coordination — developing adaptability across high-acuity, fast-paced medical environments.

**TEAS — Tippecanoe Emergency Ambulance Service | Lafayette, IN** *Fall 2026*

***Field Ride-Along Rotation | ~28 hours | Pre-Hospital Emergency Response***

- Completed a 28-hour field ride-along with the local Lafayette EMS system, responding to live emergency calls alongside certified EMTs and paramedics in a pre-hospital environment.
- Observed and assisted with patient assessment, scene management, and emergency medical interventions across a range of medical and trauma calls.
- Applied NREMT training in real-world field conditions, reinforcing clinical decision-making, patient communication, and protocol adherence under time-critical circumstances.

## TECHNICAL PROJECTS

**TraineeFlightOps (TFO) — Pilot Training Management System | Project Manager (3-person team) · Systems Development & Security · Purdue CNIT 18200** *Sep 2026 – Dec 2026*

- Serve as Project Manager for a 3-person team building TFO, a web-accessible information system tracking pilot trainee flight progress, course/lesson prerequisites, and instructor credentials for a simulated aviation-department client.
- Authored the project charter and own scope, schedule, and deliverable management, applying enterprise-grade systems development and organizational security methodologies.
- Coordinating requirements, architecture, and security-control decisions with two engineering teammates against a Dec 2026 delivery milestone.

**LostLink — Community-Driven Campus Lost & Found Platform | Team Project (5) · Figma · User-Centered Design · Purdue CS 47500 (HCI)** *Sep 2026 – Present*

- Serving as one of five team members designing LostLink, a community-driven lost-and-found platform for Purdue's campus, chosen as the team's Milestone 0 project direction.
- Applying a full user-centered design lifecycle — need-finding research, high-fidelity Figma prototyping, and planned usability testing — across a semester-long HCI project.
- Contributing to the team's project proposal defining target users, current gaps in campus lost-and-found processes, and the proposed platform approach.

**Relay — Incident & Emergency Response Coordination Platform | Independent Project · Product Management & Backend Engineering** *2026 – Present*

- Independently conceiving, designing, and building Relay, a platform that routes incidents to the nearest qualified on-duty responder across departments (EMS, security, event operations), replacing siloed, organization-based radio and roll-call workflows.
- Owning both product management and backend engineering end-to-end, applying real-world lessons from EMT field work and large-event operations to cross-department responder matching and structured incident handoff to arriving EMS.

**AeroLog — Universal Flight Log & Tracker | Next.js (App Router) · TypeScript · PostgreSQL (Neon) · Tailwind CSS · Purdue CS 348** *2026*

- Designed and built AeroLog, a database-backed web application for pilots, frequent flyers, and aviation enthusiasts to log flights they piloted, boarded as a passenger, or spotted.
- Modeled a normalized PostgreSQL schema (flights, airlines, airports, aircraft types) and implemented full CRUD workflows plus a filterable analytics/reporting interface for CS 348 Database Systems.

**Custom Agentic AI Chatbot | Python · Ollama · RAG · Prompt Engineering · CI/CD** *Dec 2025 – Present*

- Architecting an evolution from a RAG-based bot to a fully agentic AI model with memory, tool-use, and autonomous reasoning capabilities using Ollama and custom training datasets.
- Applied advanced prompt engineering, bot monitoring, and AI workers within a CI/CD framework to support iterative, test-driven model improvement.

**Wine Yield Forecasting — BASF x The Data Mine | Python · Scikit-learn · MLOps · Time-Series** *Aug 2024 – Dec 2024*

- Delivered a machine learning time-series forecasting model achieving 92% sales prediction accuracy, enabling data-driven inventory and marketing decisions for BASF.
- Translated complex stakeholder business requirements into precise technical specifications, aligning ML model scope with client goals and KPIs.
- Designed and deployed an automated MLOps pipeline reducing model update turnaround from 2–3 days to ~1–2 hours — a 97% efficiency improvement — ensuring access to current predictions in near-real time.
- Refactored and debugged legacy analytics code, improving overall system reliability by 30%.

**AI Portfolio Platform with RAG Chatbot | Next.js 15 · TypeScript · PostgreSQL · Vercel · CI/CD** *2024 – Present*

- Engineered a production full-stack AI portfolio platform with a conversational RAG chatbot using semantic search over embedded resume and project data — demonstrating applied NLP and retrieval-augmented generation.
- Configured CI/CD pipeline on Vercel with automated deployments, secure API key management, and environment variable governance following security best practices.
- Designed a serverless contact form system with PostgreSQL persistence, IP geolocation tracking, and transactional email delivery (Resend API) — capturing visitor engagement metrics for data-driven iteration.

## LANGUAGES

**English** (Fluent) • **Telugu** (Proficient) • **Hindi** (Intermediate) • **French** (Beginner)